

The AI/ML Interview Book

From Scratch to Pro — Theory, Code, and Interview Q&A

front_matter

AYuSh

Contents

Preface 3

Preface

If you are preparing for a machine learning interview in the current market, you are facing a strange paradox. The field has never been more accessible — free courses, open models, and endless tutorials — and yet interviews have never been harder to prepare for. The bar has moved. A decade ago you could pass an ML interview by explaining the bias-variance tradeoff and implementing k-means. Today you may be asked to derive backpropagation on a whiteboard at 10 a.m., design a retrieval-augmented generation system at 1 p.m., and debug a training loss curve at 4 p.m. — all for the same job.

This book exists to carry you across that entire range. It is written for three kinds of readers at once. The **complete beginner** who has decided to enter the field and wants a single, sequential path from "what is a matrix?" to "how does speculative decoding work?". The **practitioner** who uses scikit-learn or PyTorch daily but has never had to justify *why* Adam differs from RMSprop out loud, under pressure. And the **interview-ready candidate** who needs dense, honest model answers rather than hand-waving.

Three principles govern every chapter:

Nothing is assumed. Every topic begins in plain language, as if you have never met it before, and then climbs — intuition first, then formal definitions, then derivations with every symbol defined, then a small worked numeric example you can check with a pencil.

All code runs. Every code listing in this book was executed before it was printed. Where output is shown, it is real output. You can type any listing into a Python file and run it.

Interviews are the destination. Every chapter ends with real interview-style questions and model answers written at the depth an interviewer actually expects — including notes on what the interviewer is listening for.

How to use this book

The book has eight parts and is designed to be read two different ways.

The scratch-to-pro path (8-16 weeks). Read sequentially, Part I through Part VIII. Part I builds the mathematical and programming foundation; nothing later will assume more than it teaches. Do the worked examples by hand. Type out and run the code. This path turns a motivated beginner into a credible candidate.

The interview-in-two-weeks path. If you already work in the field, go straight to where roughly 80% of interview questions concentrate: Part II (classical ML), Part III (deep learning), and Part IV (NLP and LLMs), plus Chapter 34 (implement-from-scratch — the

single most requested interview practice topic). Skim the Q&A sections of every other chapter, and read Part VIII the weekend before your loop.

Whichever path you take, treat the end-of-chapter Q&A as active practice, not reading material: cover the answer, say your own answer out loud, then compare. Speaking an answer is a different skill from recognizing one.

A note on the appendices: Appendix A (formula cheat sheets) and Appendix C (100 rapid-fire one-liners) are designed for the final 48 hours before an interview. Appendix E (SQL and Python quick reference) covers the data-manipulation screens that increasingly precede ML rounds.

Notation

The same symbols mean the same things in every chapter of this book. When a chapter introduces a symbol not listed here, it defines it at first use.

Symbol	Meaning
x	a single input (feature vector), $x \in \mathbb{R}^d$
X	the design matrix of all inputs, shape $n \times d$ (rows are examples)
y	the true label or target
$\hat{y}$	the model's prediction
n	number of examples; d — number of features
w, b	weight vector and bias of a single model or layer
θ	the set of all learnable parameters of a model
η	learning rate
$\mathcal{L}$	loss function (to be minimized)
$\nabla_{\theta} \mathcal{L}$	gradient of the loss with respect to parameters
$\sigma(\cdot)$	the sigmoid function $\sigma(z) = 1/(1 + e^{-z})$
$\mathbb{E}[\cdot]$	expectation; $\text{Var}(\cdot)$ — variance
$P(A B)$	conditional probability of A given B
$\mathcal{N}(\mu, \sigma^2)$	Gaussian distribution with mean μ and variance σ^2
$D_{KL}(P \ Q)$	Kullback–Leibler divergence from Q to P
$H(P)$	entropy of distribution P
$\ v\ $	Euclidean (L2) norm of vector v

Symbol	Meaning
A^T	transpose of matrix A ; A^{-1} — inverse
λ	regularization strength (or an eigenvalue — context makes it clear)

Vectors are column vectors unless stated otherwise. Code follows Python conventions: 0-indexed, row-major NumPy arrays where $x[i]$ is the i -th example.